

Correct Fix Probability Improvement Method via INS Aiding to Single Epoch RTK System

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ABSTRACT

This paper introduces a method that utilizes an Inertial Navigation System (INS) to improve the Probability of Correct Fix (PCF) in a single-epoch Real-Time Kinematic (RTK) system. In the proposed system, only single-epoch GNSS measurements are used to avoid problems caused by cycle slips, and a time-propagated relative vector in the INS is used as a pseudo measurement in addition to the GNSS measurements to obtain an RTK solution with an improved PCF. We mathematically prove that the covariance matrix of an ambiguity float solution is always improved by the addition of prior information of an INS-aided relative vector compared to that of the existing single-epoch RTK. However, the PCF simulation results obtained by using the improved covariance matrix after applying the proposed method show that counter examples with a worse PCF of INS-aided solution than that of a GNSS-only solution may occur due to the characteristics of Z-transformation in the Least-squares AMBIGUITY Decorrelation Adjustment (LAMBDA) method. This study also mathematically proves that the PCF is always improved without a counter example when the original Z matrix calculated from the existing single-epoch RTK is applied. Thus, we propose a new algorithm that does not generate any counter examples, while increasing the PCF significantly based on the proof. Availability simulations were performed for different GNSS measurement noise levels and Inertial Measurement Unit (IMU) sensor grades. The results show that the system availability can be improved after applying the proposed method to the existing single-epoch RTK.

1. INTRODUCTION

Autonomous ground and air vehicles, dominating the world's highways and limited airspace in the future, may require significantly improved navigation systems than are currently available. The developers of autonomous vehicles are exploring different methods of obtaining sub-centimeter positioning accuracy. A Real-Time Kinematic (RTK), which is a carrier-based differential global navigation satellite system (CDGNSS), is one such method. Along with the highly accurate positioning results of the RTK system, high levels of integrity in autonomous navigation are key to implementing safety-of-life automotive applications. Integrity relates to trustworthiness of the information provided by the navigation system. An essential functionality for ensuring integrity of the information is to detect navigation faults in real time and provide integrity risk information to the users, such as the protection level bounding errors in the estimated position solutions to the required probability of hazardous misleading information ($PHMI_{req}$).

The key to high-precision RTK positioning is the process of resolving the unknown ambiguities of carrier-phase measurements as integers. The RTK system can provide precise positioning only when the integer ambiguities are correctly resolved. The ambiguity resolution procedure can be divided into two steps [1]: 1) discarding the integer nature of the ambiguities and performing a standard adjustment, such as the least-squares estimator or Kalman filter, to compute a float ambiguity solution, and 2) mapping the float ambiguity solution from its real number space to the integer number space to account for the integer nature of the ambiguities. Once the integer ambiguities are computed, an acceptance test is conducted to check whether or not the integer solution is correct. Several such tests have been proposed, and the ratio test is one of the most popular [2]. If the integer ambiguity solution passes the acceptance test, the ambiguity residual, which is the difference between the float and integer ambiguity solutions, is used to readjust the float estimate of the position. The readjusted position is referred to as the fixed position. It is important to evaluate the correctness of the integer ambiguity solution because incorrectly resolved or fixed ambiguities may cause a bias in the position of the user, which may pose a threat to the navigation integrity. The ratio test is not the test for the correctness of the fixed ambiguity solution, but for the closeness of the float ambiguity solution to its nearest integer [3]. This indicates that biases of arbitrary integer size can be present in the float ambiguity solution without ever being detected by the ratio test [3], which provides hazardous misleading information to the users. In general, the ratio-test result might be related to the Probability of Correct Fix (PCF); however, it does not guarantee a demandingly high PCF that satisfies the high integrity. Furthermore, calculating the PCF under the given ratio test results is computationally demanding, as it involves the integral equation that links the PCF to the size of the acceptance region in a real

number space [3]. Therefore, determining the acceptance of the fixed ambiguity solution based only on the ratio test is not enough for high-integrity applications.

The shipboard relative global positioning system (SRGPS) is an architectural variant of the joint precision approach and landing system (JPALS) [4]. It was developed by the US Navy to provide high-accuracy and high-integrity CDGNSS navigation for automatic shipboard landings. In previous studies [5, 6], the protection level for the fixed position error was derived, and a detailed analysis of the fault-free integrity and availability of SRGPS was conducted. The SRGPS uses integer bootstrapping (IB) to fix the ambiguities because it is possible to directly compute the PCF. In contrast to conventional RTK using the ratio test, the integer solution is accepted when the probability of incorrect fix (PIF) (i.e., one minus PCF) is smaller than the given limited integrity budget allocated to incorrect fixes (e.g., $0.1 \times PHMI_{req}$) [4]. Thus, the navigation algorithm of the SRGPS was designed to provide substantial filtering (as long as 15 min or more) to ensure the integrity of the integer ambiguity resolution [4].

The substantial filtering performs with an assumption that the global navigation satellite system (GNSS) carrier phase signals are continuously tracked without cycle slips. Because the integer ambiguity changes when a cycle slip occurs, it must be newly estimated in the ambiguity estimation filter. An SRGPS aircraft uses a high-end GPS receiver exploiting the deep integration with an inertial measurement unit (IMU) to prevent cycle slips [4], and conducts the landing within a well-controlled environment on the sea without signal blockages. However, in autonomous vehicle operations, cycle slips may frequently occur due to relatively low-cost GNSS equipment, high dynamic movement of the vehicle, and signal blockage caused by the surrounding urban environment [7]. Therefore, it is necessary to detect and fix a cycle slip when using conventional RTK systems. However, a cycle slip may not be detected due to measurement noise, which leads to incorrectly fix the integer ambiguity. This may result in a bias in the user's position, affecting the navigation integrity. A single-epoch-based RTK system can be used to avoid the effect of cycle slip causing such an integrity threat because it estimates the integer ambiguity using the GNSS measurements received at each epoch without assuming that the integer ambiguity is constant. The single epoch-based RTK has a lower PCF compared to a conventional RTK since the amount of the information (i.e. the measurements) used to estimate the ambiguities is smaller than that of the conventional RTK. Therefore, when the PCF-based acceptance test is used, the acceptance rate of the single epoch-based RTK might be poorer than that of the conventional RTK, indicating that users cannot fully exploit the carrier-phase measurements.

Ways of improving ambiguity resolution performance in RTK systems has received considerable research attention. One of the most used methods is fusing the GNSS sensor with other auxiliary sensors. The inertial-navigation-system (INS)-integrated RTK system has been investigated in several studies for applications requiring rapid and accurate on the fly integer ambiguity resolution to achieve a few centimeters' accuracy in positioning, such as [8-15]. Inertial measurement unit (IMU) measurements were used to propagate the previous position to the current position, which constrains the search space of the ambiguities. The reduced ambiguity search space significantly improves ambiguity resolution performance, especially in GNSS challenged areas [15, 16]. In [17, 18], methods were proposed to detect cycle slips by using the estimated position from the INS information to provide accurate and continuous positioning performance. All previous studies, however, are limited to improving the positioning accuracy, not the integrity. To ensure the integrity of the carrier-phase-based system, a very high level of PCF requirements must be satisfied. In [19, 20, 21], high-integrity CDGNSS techniques satisfying demanding high PCF were proposed, which are the geometry extra redundant almost fixed solutions algorithm [19], the enforced position-domain integrity risk of cycle resolution algorithm [20], and the generalized integer aperture bootstrapping algorithm [21]. These methods, however, only use the GNSS measurement for the ambiguity resolution without other auxiliary sensors. This study proposes the use of INS to achieve high levels of integrity. Our approach is based on the use of the time updated position from the INS to maximize the PCF. Our proposed method provides a significant improvement in PCF, and it is mathematically proved that the PCF always increases regardless of the noise level of the INS. The performance of the proposed method was evaluated in the protection level domain with respect to the grade of the IMU sensors and the GNSS measurement error models.

This paper is organized as follows. In Section 2, the system architecture that utilizes the INS information to improve the PCF in the LCKF-based single-epoch RTK/INS system is described. The decrease of estimation uncertainty is mathematically proved, and the PCF simulation results of the proposed architecture are presented. The PCF improves at most of epochs, but it is also confirmed that counterexamples occur in which the PCF value is lowered after INS aiding. Section 3 discusses the Z-transformation which causes the counterexamples due to its numerical computation process. A Z_0 -Based ambiguity decorrelation method which increases the PCF at every epoch without any is proposed. A Z-matrix selection method maximizes the improvement of PCF by combining the conventional Z transformation method and the Z_0 -Based ambiguity decorrelation method. Finally, in Section 4, the performance of the proposed method is evaluated in the protection level domain with respect to the grade of the IMU sensors and the GNSS measurement error models.

2. INS-INTEGRATED SINGLE-EPOCH RTK

2.1. System Architecture

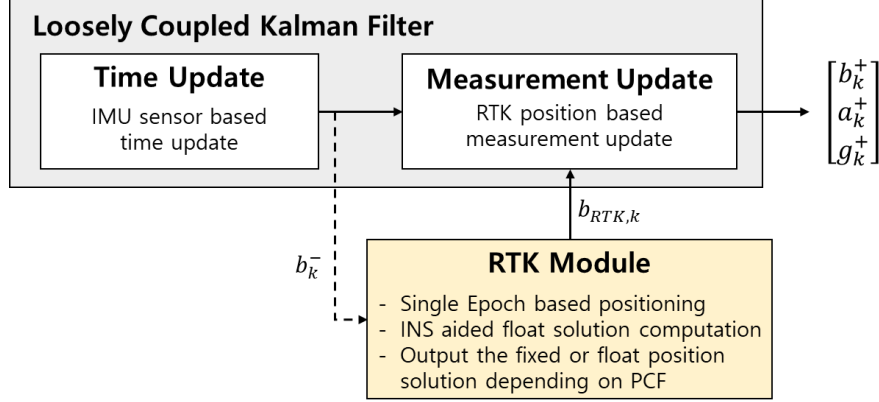


Fig. 1. Schematic diagram of the system architecture of the RTK/INS system.

In this study, an LCKF-based RTK/INS system was used to integrate the GNSS and IMU sensors, as shown in Fig. 1. The state of the LCKF is composed of a position (i.e. a relative vector between the two receivers) which is represented as b in Fig. 1, and inertial sensor biases which are represented as a for the accelerometer and g for the gyroscope in Fig. 1. The LCKF state equations at any discrete time k are described by a time update (Equation (1)) and a measurement update (Equation (2)) [22]. In Equation (1), the IMU sensor measurements $u_{IMU} = [f_{ib}^b, \omega_{ib}^b]^T$ are used as a control input in the time update step, which is a 6×1 vector (three-axis measurements for an accelerometer f_{ib}^b and a gyroscope ω_{ib}^b). In Equation (2), the position computed at the RTK module is used as a measurement input in the measurement update step to correct the INS-propagated state.

$$x_k^- = F(u_{IMU})_k \cdot x_{k-1}^+ \quad P_k^- = F(u_{IMU})_k \cdot P_{k-1} \cdot F(u_{IMU})_k^T + W_k \quad (1)$$

$$x_k^+ = x_k^- + K_k(z_{RTK} - A_k x_k^-) \quad P_k^+ = (I - K_k A_k) P_k^- \quad (2)$$

In Equations (1) and (2), x_k^- and x_k^+ are the time updated and measurement updated state vectors at time k , respectively. P_k^- and P_k^+ are the covariance matrices of the x_k^- and x_k^+ , respectively, and z_{RTK} is the measurement vector (output of the RTK module in this study). $F(u_{IMU})_k$ and A_k are the state transition matrix as a function of the IMU sensor measurements and measurement model matrix, respectively. K_k is the Kalman gain matrix, W_k is the covariance matrix of the process errors, and R_k is the covariance matrix of the measurement error (output of the RTK module in this study).

The RTK module of the proposed system uses a least-squares filter based on the Double Differenced (DD) GNSS measurements to estimate the float ambiguity and float position solution. The float ambiguity solution is used to estimate the integer ambiguity solution. The RTK module outputs either the float position or fixed position solution according to the acceptance test results. In the proposed method, the single-epoch DD GNSS measurements are used to avoid the problem caused by cycle slips. However, the PCF of the single-epoch based RTK is lower than the conventional RTK which estimates the integer ambiguities more accurately by using the GNSS raw measurements from multiple sequential epochs. Furthermore, in a general LCKF structure, the INS system cannot be utilized in the ambiguity resolution process because the ambiguities are estimated at the external RTK filter and not at the RTK/INS Kalman filter. Thus, we used the time-updated position from the INS as the pseudo measurement b_k^- in addition to the DD GNSS measurements to improve the PCF in the RTK module. The least-square estimates of the float ambiguity and the float position solution can be calculated as follows:

$$\begin{bmatrix} \nabla \Delta \rho_k \\ \nabla \Delta \phi_k \\ b_k^- \end{bmatrix} = H \begin{bmatrix} b \\ \nabla \Delta N \end{bmatrix} + \varepsilon \quad \text{where } H = \begin{bmatrix} \nabla G & 0 \\ \nabla G & \Lambda \\ I & 0 \end{bmatrix}. \quad (3)$$

$$\begin{bmatrix} \hat{b}_{LS} \\ \nabla \Delta \hat{N}_{LS} \end{bmatrix} = S \begin{bmatrix} \nabla \Delta \rho_k \\ \nabla \Delta \phi_k \\ b_k^- \end{bmatrix} \quad (4)$$

where $b \in \mathbb{R}^{3 \times 1}$ and $\nabla \Delta N \in \mathbb{R}^{n \times 1}$ are the true position and DD ambiguities, respectively. $\nabla \Delta \rho_k \in \mathbb{R}^{n \times 1}$ and $\nabla \Delta \phi_k \in \mathbb{R}^{n \times 1}$ are the DD code and carrier phase measurements, respectively, and b_k^- is the time-updated position at the Kalman filter. $\hat{b}_{LS}$ and $\nabla \Delta \hat{N}_{LS}$ are the float position solution and the float ambiguity solution, respectively. ∇G and Λ are the satellite differenced geometry and the wave length matrices, respectively, and S is the pseudo-inverse matrix of H and ε is the measurement noise.

Once the float ambiguity solution is obtained, it is mapped into the integer number space to consider the integer constraints of the ambiguities. The widely known methods to map a float ambiguity solution to an integer solution include integer rounding (IR), integer bootstrapping (IB), and integer least squares (ILS). In this study, the float ambiguity solution was mapped into the integer solution by the IB. In contrast to IR or ILS, the IB method enables us to compute the PCF in a straightforward manner, which will be described in Section 2.2 [1].

After obtaining the integer solution of the ambiguities, which is referred to as a fixed ambiguity solution, the acceptance of the fixed ambiguity solution must be decided. In this study, the fixed ambiguity solution was accepted according to the PIF (i.e., $1 - \text{PCF}$), which is calculated from the variance–covariance matrix of the float solution, as described in Section 2.2. When the computed PIF is smaller than the given integrity risk allocated for the fault-free condition ($PHMI_{req,0}$), the fixed ambiguity solution is accepted, and the solution is used to correct the float position. The rationale behind the PCF (or PIF)-based acceptance test is that a fixed ambiguity solution with a low PIF is sufficiently reliable to ensure the required integrity. As a result, the ambiguity-resolved position ($\check{b}$), usually referred to as a fixed position, can be obtained, and the RTK module outputs the RTK position $\check{b}$ (which will be an input to LCKF), as shown in Equation (5). On the other hand, when the computed PIF is larger than $PHMI_{req,0}$, the RTK module outputs the float position solution, $\hat{b}_{LS}$, as the RTK position.

$$b_{RTK} = \begin{cases} \hat{b}_{LS} & \text{if } PIF \geq PHMI_{req,0} \\ \check{b} & \text{if } PIF < PHMI_{req,0} \end{cases} \quad (5)$$

2.2. Z-transformation and Probability of Correct Fix of Integer Bootstrapping

In this section, we briefly review and summarize the previous papers related to the Z-transformation [23] and PCF of the IB [1, 24, 25].

2.2.1. Z-transformation

The entries of the float ambiguity solution are strongly correlated, and their confidence ellipsoids are extremely elongated [1] because the DD GNSS measurements, used to compute the float ambiguity solution, share a common satellite measurement. This large correlation disturbs the efficient estimation for a fixed ambiguity solution. The Z-transformation decorrelates the float ambiguity solution, which makes its variance–covariance matrix diagonalized. The variance–covariance matrix of the Z-transformed float ambiguity solution is expressed as follows [25]:

$$Q_{\nabla \Delta N_Z} = Z^T \cdot Q_{\nabla \Delta N} \cdot Z \quad (6)$$

where $Q_{\nabla \Delta N_Z}$ is the covariance matrix of the Z-transformed float ambiguity solution and Z is the Z-transformation matrix. Although the aim of the Z-transformation is to decorrelate the ambiguities, the diagonality of the variance–covariance matrix is difficult to attain [24]. This is because the Z-matrix must satisfy the constraints that both Z and its inverse matrices have integer entries and the determinant of Z is equal to +1 or −1, while minimizing the diagonal entries of the variance–covariance matrix [24]. Thus, there is no analytical closed-form expression for the Z-matrix, and it must be computed numerically.

2.2.2. Probability of Correct Fix of Integer Bootstrapping

After Z-transforming the float ambiguity solution, it is mapped into the integer number space to consider the integer constraints of the ambiguities by using the IB. The IB method is a generalization of the IR method considering the correlations between the

estimated ambiguities, while IR simply treats the multivariate integer estimation problem as a problem with n independent scalar estimation problems [21]. This follows sequential conditional least-squares adjustment, which can be divided into two steps: 1) rounding the k -th real valued ambiguity to the nearest integer, and 2) correcting the $k + 1$ to n -th remaining real valued ambiguities by virtue of their correlation with the k -th ambiguity where k is from 1 to n (n is the number of the ambiguities) [1].

Since the correction step fully decorrelates the float ambiguities, the PCF of the IB equals the product of PCF of each ambiguity [25], which enables us to compute the PCF easily and exactly in a straightforward manner. The probability that the i -th ambiguity is correct fix is calculated by integral of the probability density function (PDF) of the conditional least-squared float ambiguity solution over $[-\frac{1}{2}, \frac{1}{2}]$. The connection of IB with triangular decomposition of $Q_{\nabla\Delta N_Z}$ is useful to determine its PDF, and thus we have the following statistical interpretation of the entries of L and D [1]:

$$Q_{\nabla\Delta N_Z} = L \cdot D \cdot L^T \quad (7)$$

where L_{ij} represents the correlation coefficient between the i -th float ambiguity and the j -th conditional least-squared float ambiguity. D_{ii} represents the standard deviations of the i -th conditional least-squared float ambiguity. Once the D matrix is obtained, the PCF or PIF can be calculated as follows [25]:

$$P(CF) = \prod_{i=1}^n \left(2\Phi\left(\frac{1}{2D_{ii}}\right) - 1 \right) \quad (8)$$

$$P(IF) = 1 - P(CF) \quad (9)$$

where $P(CF)$ and $P(IF)$ are the probability of correct fix and incorrect fix, respectively. Φ is the cumulative density function of the standard normal distribution.

2.3. PCF of the INS-Integrated Single-Epoch RTK

The estimation uncertainty can be decreased by using the time-updated position from the INS as the pseudo measurement in addition to the GNSS measurements to estimate the float ambiguity solution. This section provides a mathematical description of the benefits of the proposed method. For the sake of convenience, hereafter, the float ambiguity solution estimated using the time-updated position from the INS as pseudo measurements is referred to as the INS-aided float ambiguity solution.

The variance–covariance matrix of the INS-aided float ambiguity solution ($Q_{\nabla\Delta N}^{INS}$) can be expressed as a function of the variance–covariance matrix of the pseudo measurement from the INS. Its derivation can be found in Appendix A and the results are as follow:

$$Q_{\nabla\Delta N}^{INS} = \Lambda^{-1} \left\{ Q_{\nabla\Delta\phi} + \nabla G \cdot (Q_{b_{KF}}^{-1} + \nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G)^{-1} \cdot \nabla G^T \right\} \Lambda^{-1} \quad (10)$$

where $Q_{b_{KF}}$ is the variance–covariance matrix of the time-updated position. In this study, $Q_{b_{KF}}$ is a 3 x 3 submatrix corresponding to the position of the time-propagated covariance matrix P_k^- . $Q_{\nabla\Delta\phi}$ and $Q_{\nabla\Delta\rho}$ are the variance–covariance matrices of the DD carrier and code measurement errors, respectively. If the time-updated position from the INS is not used as the pseudo measurement (i.e., $Q_{b_{KF}} \rightarrow \infty$), the variance–covariance matrix of the float ambiguity solution ($Q_{\nabla\Delta N}^0$) can be expressed as follows:

$$Q_{\nabla\Delta N}^0 = \Lambda^{-1} \{ Q_{\nabla\Delta\phi} + \nabla G \cdot (\nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G)^{-1} \cdot \nabla G^T \} \Lambda^{-1} \quad (11)$$

For convenience, hereafter, the float ambiguity solution estimated without the pseudo measurement is referred to as the GNSS-only float ambiguity solution. Appendix B shows that the matrix obtained by subtracting $Q_{\nabla\Delta N}^{INS}$ from $Q_{\nabla\Delta N}^0$ is a positive semi-definite matrix, and can be expressed as follows:

$$v^T \cdot (Q_{\nabla\Delta N}^0 - Q_{\nabla\Delta N}^{INS}) \cdot v \geq 0 \quad \text{for } \forall v \in \mathbb{R} \quad (12)$$

One of the properties of the positive semi-definite matrix is that its diagonal entries are nonnegative. Thus, the diagonal entries of

$Q_{\nabla\Delta N}^0$ are larger than those of $Q_{\nabla\Delta N}^{INS}$, which implies that the variances of the float ambiguities decrease when the time-updated position is used as a pseudo measurement. The reduced uncertainty of the float ambiguity solution can increase the PCF.

A more rigorous proof considering the Z-transformation and LDU decomposition steps is required to show the improvement of the PCF. However, this is a highly challenging task because of the numerical computation process of the Z-matrix. In this section, the PCF was simply simulated by assuming a constant time-updated position error uncertainty to verify the improvement of the PCF. For the simulation, $Q_{b_{KF}} = \sigma_{b_{KF}}^2 \cdot I_{3 \times 3}$ is used, where $\sigma_{b_{KF}}$ is a constant standard deviation of the pseudo measurement error. A dual constellation and dual frequency (DCDF) scenario was used for 24 hours at an interval of one second. The GNSS constellation and error models are summarized in Table 1.

Table 1. Sensor error model for the simulation

GNSS constellation	24 GPS satellites [26] / 24 Galileo satellites [27]
Frequency	GPS: L1 & L5 Galileo: E1 & E5
GNSS multipath error and noise model	LAAS AAD-B [28]
$\sigma_{b_{KF}}$	0.1 m / 0.5 m

Table 2 shows a significant improvement in the PCF (i.e., a significant decrease in PIF) when the pseudo measurement was used from the INS. Because the PCF is close to one, it is difficult to compare the results clearly. For the sake of convenience, hereafter, the results are discussed in the PIF domain. Table 2 shows the average of the PIF values (μ_{PIF}) with respect to the standard deviation of the pseudo measurement error ($\sigma_{b_{KF}}$). “No INS” is for the case of the GNSS-only float ambiguity solution. Even when $\sigma_{b_{KF}}$ was conservatively set to 0.5 m, μ_{PIF} was decreased by 31.6% compared to the “No INS” case. When $\sigma_{b_{KF}}$ was set to 0.1, it further decreased μ_{PIF} by 88.9% compared to the “No INS” case. However, we observed some counter examples for which the PIF of the INS-aided float solution is higher than that of the GNSS-only case, as shown in Fig. 2. Counter examples occurred at 13,171 epochs out of 86,400 epochs when $\sigma_{b_{KF}}$ was 0.5 m, and at 1,445 epochs when $\sigma_{b_{KF}}$ was 0.1 m. Because the proposed method aims at improving the PCF to guarantee a high level of navigation integrity, counter-examples are undesirable. Therefore, in the following section, we discuss the factors that increase the PIF and propose a method to decrease the PIF without counter examples.

Table 2. PIF results with respect to the uncertainty of the time-updated position at the Kalman filter.

$\sigma_{b_{KF}}$	μ_{PIF}
No INS	1.35×10^{-7}
0.5 m	9.23×10^{-8}
0.1 m	1.50×10^{-8}

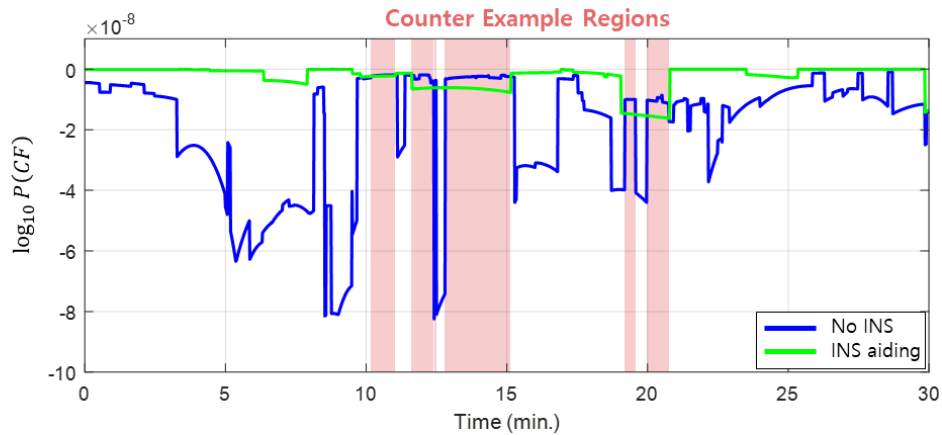


Fig. 2. Counter-example whose PCF is smaller than that of the “No-INS” case

3. PCF MAXIMIZATION METHOD OF INS-AIDED SINGLE-EPOCH RTK

3.1. Z_0 -Based Ambiguity Decorrelation Method

As described in Section 2.3, the variances of the INS-aided float ambiguities (the diagonal entries of $Q_{\nabla\Delta N}^{INS}$) always decrease compared to those of the GNSS-only (the diagonal entries of $Q_{\nabla\Delta N}^0$). On the other hand, the correlation between the float ambiguities (the non-diagonal entries of $Q_{\nabla\Delta N}^{INS}$) could increase or decrease based on the satellite geometry, as shown in Equation (10), even when the pseudo measurements are used. When the decrease in the diagonal entries of $Q_{\nabla\Delta N}^{INS}$ is relatively small (i.e., the standard deviation of the pseudo measurements is comparatively large), the variation of the non-diagonal entries might disturb the numerical computation process of the Z-matrix, which results in a worse decorrelation of the ambiguity float solution than the GNSS-only case. As a result, the PIF can be increased compared to that of the GNSS only case. This is because it is difficult to find an ideal Z-matrix that optimally decorrelates the ambiguities for a given variance–covariance matrix, as described in Section 2.2. Thus, even if the variances of the INS-aided float ambiguities are decreased, the PIF could be higher than that of the GNSS-only case. To eliminate all such counter examples, this study proposes a technique that uses the Z-matrix calculated from the variance–covariance matrix of the GNSS-only float ambiguity solution ($Q_{\nabla\Delta N}^0$), and not that of the INS-aided float ambiguity solution ($Q_{\nabla\Delta N}^{INS}$). This is used to decorrelate the INS-aided float ambiguity solution for the Z-transformation step. Hereafter, this technique is referred to as the Z_0 -based decorrelation.

It is possible to mathematically prove the improvement of the PCF by showing the reduced variance of the conditional least-square float ambiguity solution, when the Z_0 -based decorrelation is used. The decorrelated variance–covariance matrices of the INS-aided float ambiguity solution, when using the Z_0 based decorrelation, is expressed as:

$$Q_{\nabla\Delta N_z}^{INS,0} = Z_0^T \cdot Q_{\nabla\Delta N}^{INS} \cdot Z_0 \quad (13)$$

The decorrelated variance–covariance matrix of the GNSS-only float ambiguity solutions, for the comparison, is as follows:

$$Q_{\nabla\Delta N_z}^0 = Z_0^T \cdot Q_{\nabla\Delta N}^0 \cdot Z_0 \quad (14)$$

where Z_0 is the Z-matrix computed from $Q_{\nabla\Delta N_z}^0$, $Q_{\nabla\Delta N_z}^0$ is the decorrelated variance–covariance matrix of the GNSS-only float ambiguity solution, and $Q_{\nabla\Delta N_z}^{INS,0}$ is the Z_0 -based decorrelated covariance matrix of the INS-aided float ambiguity solution. Because Z_0 is full rank, the matrix obtained by subtracting $Q_{\nabla\Delta N_z}^{INS,0}$ from $Q_{\nabla\Delta N_z}^0$ is a positive semi-definite matrix as follows:

$$v^T \cdot (Q_{\nabla\Delta N_z}^0 - Q_{\nabla\Delta N_z}^{INS,0}) \cdot v \geq 0 \quad \text{for } \forall v \in \mathbb{R} \quad (15)$$

As shown in Appendix C, the PCF computed from $Q_{\nabla\Delta N_z}^0$ is greater than that of $Q_{\nabla\Delta N_z}^{INS,0}$ without counter examples when the condition of Equation (15) holds true. Thus, when the Z_0 -based decorrelation is applied, the PCF always increases.

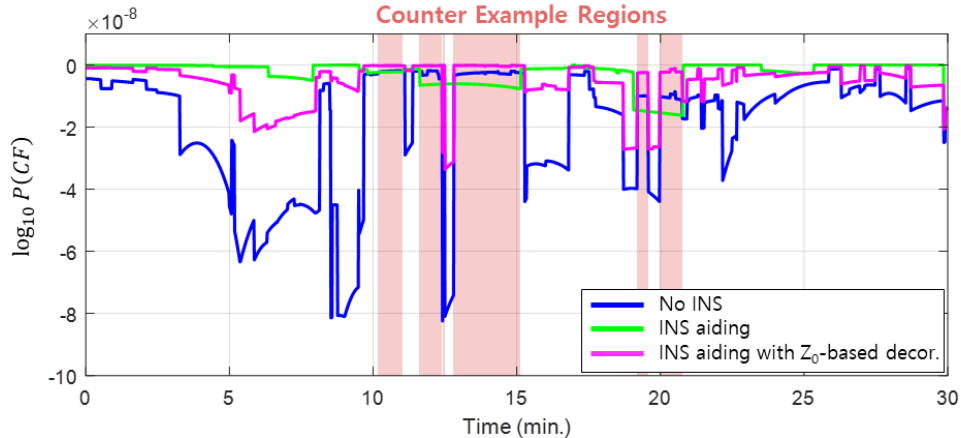


Fig. 3. PCF simulation results from the proposed Z_0 based decorrelation method.

Fig. 3 compares the three PIF simulation results: 1) the GNSS-only case (blue line), 2) the INS-aided case where the conventional Z-transformation is used (green line), and 3) the INS-aided case where the Z_0 -based decorrelation is used (magenta line). Fig. 3 shows that the PIF of the INS-aided case with the Z_0 -based decorrelation (magenta line) is always lower than that of the GNSS-only case (blue line) with no exception.

Table 3 shows the average values of the PIF (μ_{PIF}) for the INS-aided cases, in which the proposed Z_0 -based decorrelation technique is either used or not used. The average PIF is lowered when conventional Z-transformation is used, even though counter examples occur. In contrast, the proposed Z_0 -based decorrelation technique does not cause any counter examples, but the average PIF improvement is lower than the conventional Z-transformation technique. Thus, in the following section, we propose the Z-matrix selection method to minimize the average PIF without counter examples by using both techniques.

Table 3. PIF simulation results

$\sigma_{b_{KF}} \text{ (m)}$	μ_{PIF}	
	Conventional Z-transformation	Z_0 based decorrelation
No INS	1.35×10^{-7}	
0.5	9.23×10^{-8}	1.03×10^{-7}
0.1	1.50×10^{-8}	4.03×10^{-8}

3.2. PCF Maximization Method

Fig. 4 describes the Z-matrix selection method to minimize the PIF (or maximize the PCF) while eliminating counter examples. First, the variance–covariance matrices of the GNSS-only ($Q_{\nabla\Delta N}^0$) and INS-aided ($Q_{\nabla\Delta N}^{INS}$) float ambiguity solutions are computed. Then, the Z-matrix for the respective variance–covariance matrix, which is Z_0 from the $Q_{\nabla\Delta N}^0$ and Z from the $Q_{\nabla\Delta N}^{INS}$, is determined. Next, the $Q_{\nabla\Delta N}^{INS}$ is decorrelated by using Z_0 and Z , respectively, and their corresponding correct fix probability is computed. $P(CF)_0$ is the PCF when the Z_0 -based decorrelation technique is used, and $P(CF)$ is the PCF when the conventional Z-transformation method is used. The final step is to select the Z-matrix that provides a larger correct fix probability.

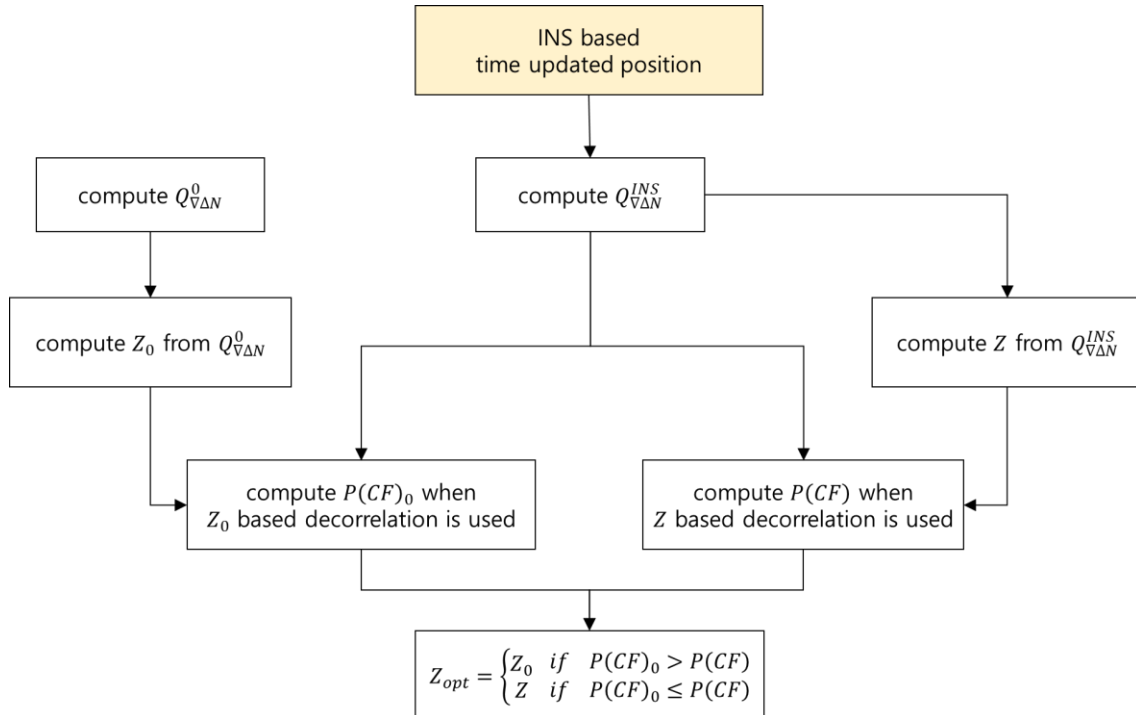


Fig. 4. Z-matrix selection method to maximize the PCF.

4. PERFORMANCE EVALUATION

In this section, the simulation results are presented to evaluate the performance of the method developed in Section 3. The performance was analyzed in the vertical protection level (VPL) domain under the fault-free condition (H_0). When the fixed ambiguity solution is not accepted at the RTK module (i.e. the “float” position from the RTK module is fed into the LCKF), the VPL can be evaluated as follows [6]:

$$VPL_{H_0}(k) = \Phi^{-1}\left(\frac{PHMI_{req,0}}{2}\right) \cdot \sigma_v \quad (16)$$

where $VPL_{H_0}(k)$ is the vertical protection level at the k -th epoch under the fault-free condition. $\Phi^{-1}(\cdot)$ is the inverse cumulative density function of the standard normal distribution. $\sigma_v(k)$ is the (3,3) entry of P_k^+ and represents the standard deviation of the vertical position error of the Kalman Filter position output. When the fixed ambiguity solution is accepted at the RTK module (i.e. the “fixed” position from the RTK module is fed into the LCKF), the VPL can be evaluated as follows [6]:

$$VPL_{H_0} = k \left(\frac{PHMI_{req,0} - P(IF)}{2 \cdot P(CF)} \right) \cdot \sigma_v \quad (17)$$

where $P(CF)$ and $P(IF)$ are the probabilities of correct fix and incorrect fix under the fault-free condition, respectively. It is noteworthy that the $P(IF)$ must be smaller than $PHMI_{req,0}$ for computing the VPL.

The VPLs are compared with the Vertical Alert Limit (VAL) to evaluate the availability of the system, which is defined as:

$$Availability (\%) = \frac{Time(VPL \leq VAL)}{Total Time} \times 100 \quad (18)$$

When the fixed position is used for the measurement update step of the LCKF, the σ_v is much smaller than that of the case where the float position is used, resulting in smaller VPLs. Thus, the availability will be highly related to the fixed ambiguity solution acceptance rate. The simulation parameters are listed in Table 4. σ_ρ and σ_ϕ represent the standard deviations of code and carrier phase measurement errors, respectively. The scaled version of the AAD-B model by using σ_ρ at a zenith direction as a reference was used for the code measurement error model. σ_ρ at an elevation angle of 90 degrees is approximately 17 cm for the AAD-B model, and the AAD-B model was modified for different σ_ρ values of 15 ~ 25 cm at a zenith direction as shown in Table 4. For σ_ϕ , a constant error model was used, regardless of the elevation angles.

Table 4. Simulation parameters for the simulation.

Parameter		Value
Requirements	$PHMI_{req,0}$	10^{-7}
	VAL (m)	0.5
IMU sensor	Grade	Consumer / Tactical
	Update rate	200 Hz
GNSS sensor	σ_ρ (cm)	15 - 25
	σ_ϕ (mm)	6 / 7 / 8
	Update rate (Hz)	1

4.1. Availability Regarding the INS Grade

The uncertainty of the pseudo measurement used to estimate the float ambiguity and position solutions, as described in Section 2.1, depends on the grades of the IMU sensor which are classified into aviation, tactical, and consumer grades. In this study, we compared the simulation results of tactical- and consumer-grade IMUs, and their noise levels are listed in Table 5.

Table. 5. Tactical-/consumer-grade IMU noise levels used for the simulation [22].

IMU sensor							
Accelerometer				Gyroscope			
Bias PSD (m^2/s^5)		Noise PSD (m^2/s^3)		Bias PSD (rad^2/s^3)		Noise PSD (rad^2/s)	
Tactical	Consumer	Tactical	Consumer	Tactical	Consumer	Tactical	Consumer
10^{-7}	10^{-5}	0.002^2	0.2^2	$2 \cdot 10^{-12}$	$4 \cdot 10^{-11}$	0.0001^2	0.01^2

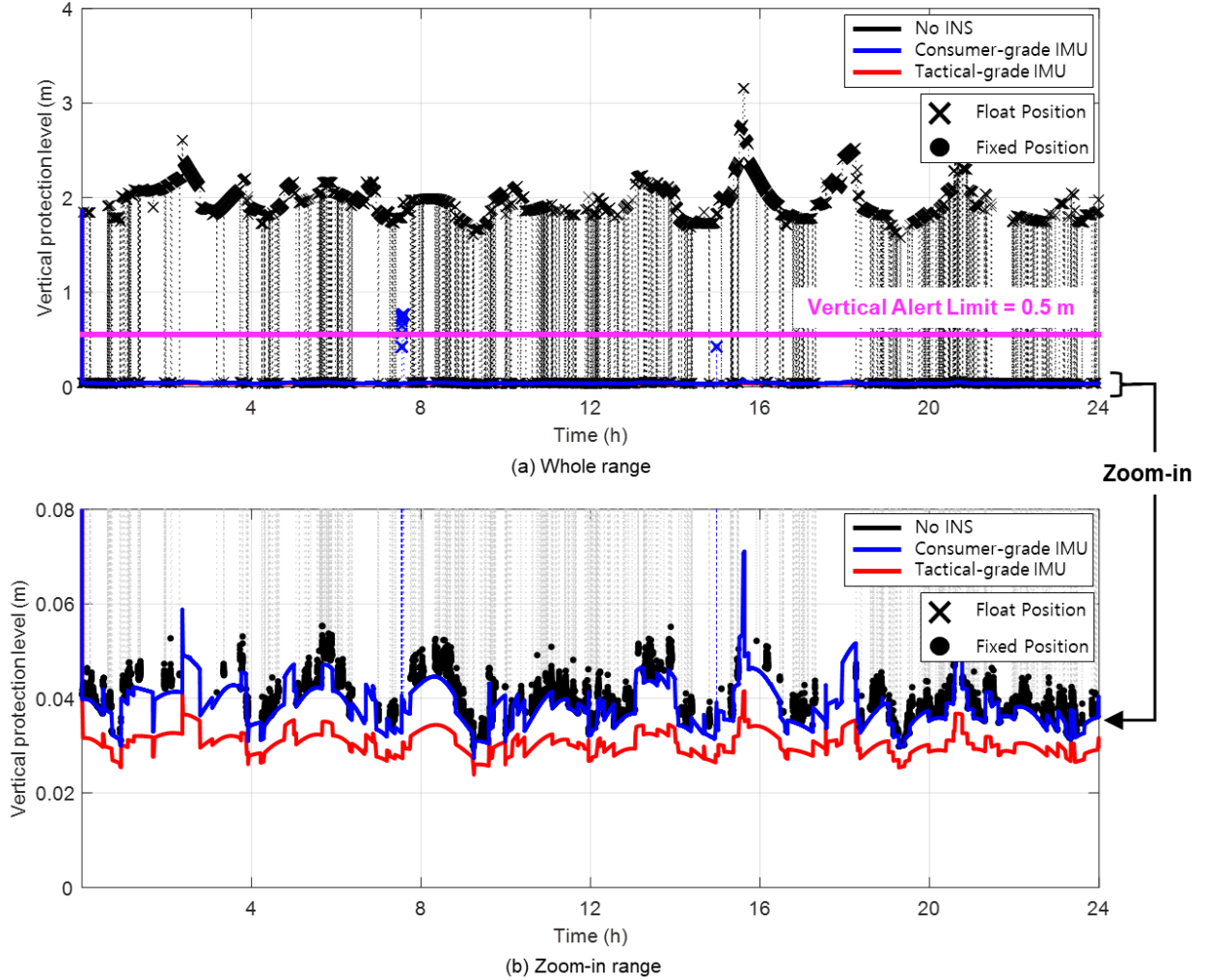


Fig. 5. VPL simulation results when $\sigma_p = 25$ cm and $\sigma_\phi = 6$ mm.

Fig. 5 shows the three VPL simulation results; 1) the black line represents the no INS case; 2) the blue line represents the consumer-grade IMU case; and 3) the red line represents the tactical-grade IMU case when $\sigma_p = 25$ cm and $\sigma_\phi = 6$ mm. The VPLs for the float and fixed cases are plotted with $\times$ markers and dot markers, respectively. In the no INS case, fixed ambiguity solutions are not accepted in many epochs because of their low PCF which results in the low fix rate. Because the VPL is lower than the VAL only when a fixed solution is available, the fixed rate and availability are equal to 32.97%.

As shown in Fig. 5, the VPLs for the float position were significantly reduced when the IMU sensor was used. The fix rates of the

consumer and tactical-grade case were 99.82% and 99.99%, respectively. Fig. 5 also shows the significant improvement of the VPL of float cases. The float VPLs of the no INS case were about 2 m in average, while that of the consumer-grade IMU case were about 0.5 m. This is because the time-updated position from the INS improves the accuracy of the float position solution as well as the PCF. The availabilities of the consumer and tactical-grade IMU cases were 99.83% and 99.99%, respectively.

4.2. Availability Regarding the GNSS Measurement Error Model

The availability simulation was performed according to the noise level of the GNSS measurement shown in Table 4 using a consumer-grade or tactical-grade IMU sensor. Fig. 6 and Fig. 7 show the availability and fix rate results, respectively, for the GNSS-only (a), the consumer-grade IMU (b), and the tactical-grade IMU (c) cases. These results indicate that the fixed rate is more dependent on the noise level of the carrier-phase measurement than that of the code-phase measurement. Furthermore, the fixed rate was significantly enhanced when the INS Aiding was applied. When the proposed method was applied using a tactical-grade IMU, the fixed rate and availability were found to be over 99.99% in all simulation ranges.

In the GNSS-only and consumer-grade IMU cases, when a float solution was used, the VPL exceeded the VAL; thus, the availability was almost the same as the fixed rate. When σ_ϕ was 6 mm, the fixed rate of the GNSS only case had a minimum value of 32.97 % and a maximum value of 92.26 %. As a result of applying the proposed method by using a consumer-grade IMU, the fixed rate was found to be over 99.99 % for all pseudo-range noise levels. When σ_ϕ was 7 mm, the fixed rate of the GNSS only case had a minimum value of 0 % and a maximum value of 9.5 %, and by applying the proposed method, it was increased to 45.17 % and 88.34 %, respectively. When the σ_ϕ was 8mm, the fixed rate of the GNSS only case was 0 % in all pseudo-range noise levels, and even if the technique was applied with a consumer-grade IMU, the maximum value was as high as 9.5 %.

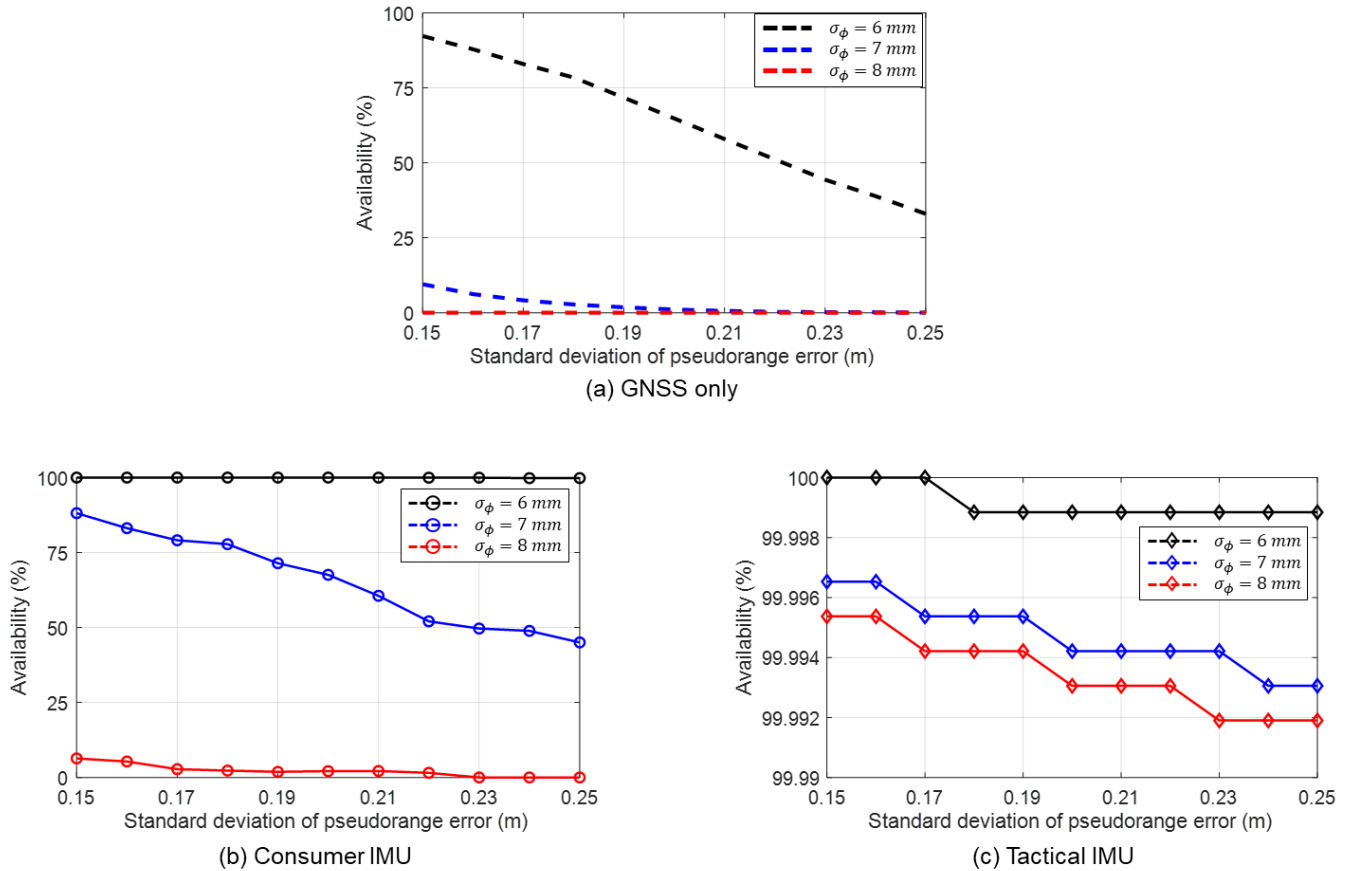


Fig. 6. Availability simulation results with respect to the GNSS measurement error models

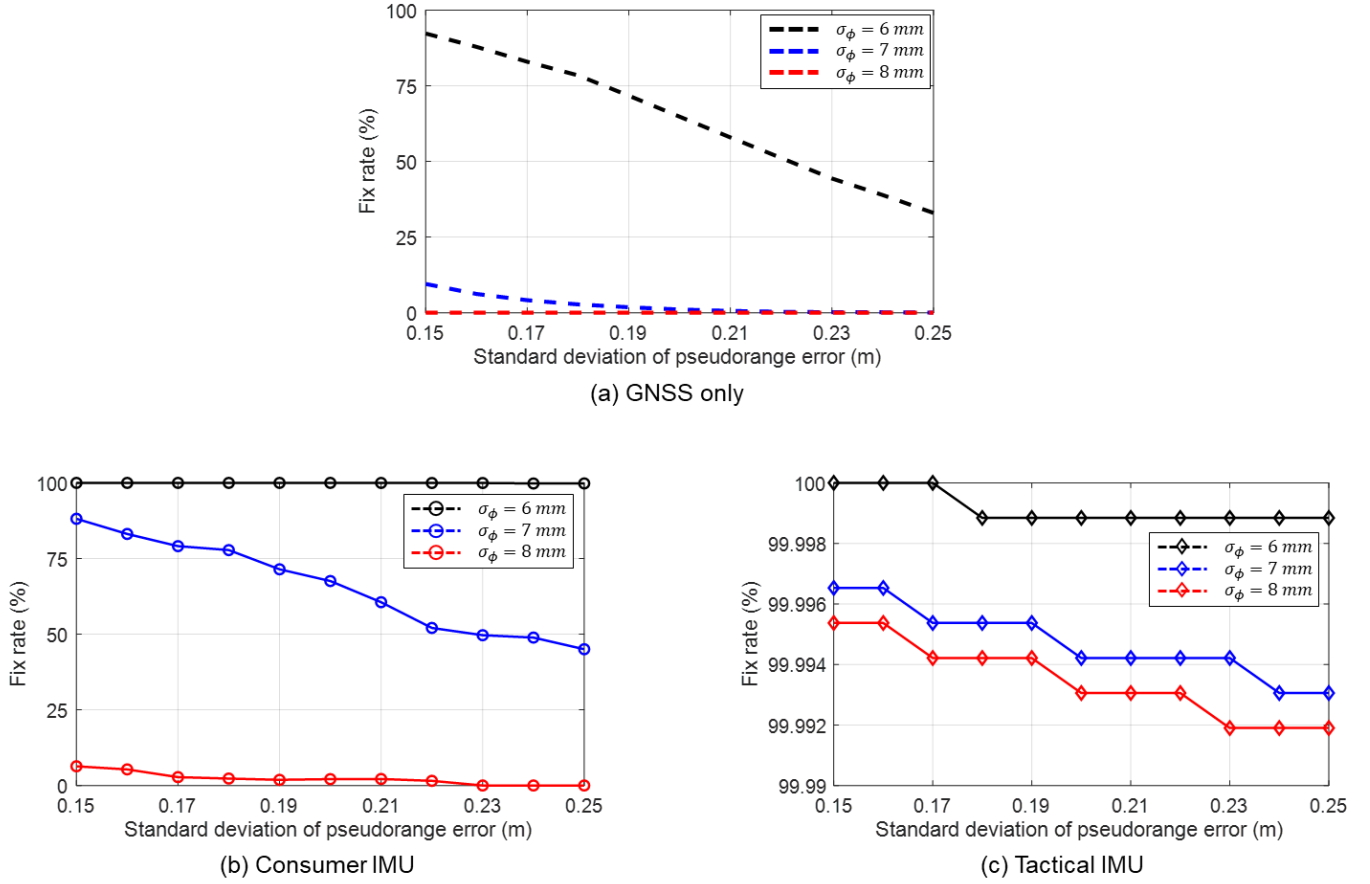


Fig. 7. Fix rate simulation results with respect to the GNSS measurement error models

CONCLUSIONS

Herein, we proposed a method that utilizes INS information to improve the PCF in a single-epoch RTK system. The covariance matrix of the float solution was derived mathematically when additional relative vector prior information was provided in the process of ambiguity resolution. In addition, we proved that the covariance matrix aided by INS is always improved when compared to that of the existing single-epoch RTK.

The results of the PCF simulation showed that INS aiding helps improve the mean of PCF, while some counter examples may occur due to the characteristics of Z-transformation. We mathematically proved that PCF is always improved without a counter example when the Z matrix calculated from the existing single-epoch RTK is applied. Finally, we proposed the method that does not generate counter examples, while maximizing the PCF improvement by selecting the optimal Z matrix.

The proposed LCKF INS-integrated single-epoch RTK was implemented, and simulations were performed for different sensor noise levels. In the case of a consumer-grade IMU, the fixed rate shows large variation according to the noise level of the GNSS measurement. When the standard deviation of the carrier-phase measurement errors was 8 mm, the fixed rate and availability were below 10 %, even if the proposed method was applied. On the other hand, in the case of using a tactical-grade IMU, the fixed rate and availability were increased to over 99.99% for all simulation ranges.

In this study, a technique for improving the PCF of single-epoch RTK was proposed, and the effectiveness of the proposed technique was verified through both mathematical proof and a simulation. The proposed method can be used for a carrier-phase-based navigation system which requires meeting the high levels of integrity and availability. Future work will compare the performance of the proposed method with those of the existing methods through real-time data processing under high dynamic flight conditions using a small drone.

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APPENDIX A. VARIANCE–COVARIANCE MATRIX OF INS AIDED SINGLE EPOCH RTK

The linear observation equation for the INS integrated single epoch RTK is given by:

$$y = Hx + \varepsilon \quad (\text{A-1})$$

where $y = [\nabla\Delta\rho, \nabla\Delta\phi, b_{KF}^-]^T$ is the measurement vector which consists of double-differenced code measurements ($\nabla\Delta\rho$), double-differenced carrier measurements ($\nabla\Delta\phi$), and time updated-position at Kalman Filter (b_{KF}^-). $x = [b, \nabla\Delta N]^T$ is the state vector composed of position (b), and double-differenced ambiguities ($\nabla\Delta N$). H is the observation matrix defined as:

$$H = \begin{bmatrix} \nabla G & 0 \\ \nabla G & \Lambda \\ I & 0 \end{bmatrix} \quad (\text{A-2})$$

where ∇G and Λ are the satellite differenced geometry and the wave length matrices, respectively. ε is the measurement noise vector, and follows a zero mean Gaussian distribution with variance–covariance Q_y :

$$Q_y = \begin{pmatrix} Q_{\nabla\Delta\rho} & 0 & 0 \\ 0 & Q_{\nabla\Delta\phi} & 0 \\ 0 & 0 & Q_{b_{KF}^-} \end{pmatrix} \quad (\text{A-3})$$

where $Q_{\nabla\Delta\rho}$, $Q_{\nabla\Delta\phi}$, and $Q_{b_{KF}^-}$ are the variance–covariance matrices of $\nabla\Delta\rho$, $\nabla\Delta\phi$, and b_{KF}^- , respectively. Then, the variance–covariance matrix of the Least-Square (LS) estimate ($\hat{x}_{LS}$) of x is given as:

$$\begin{aligned} Q_{\hat{x}_{LS}}^{INS} &= \begin{bmatrix} Q_b^{INS} & Q_{b,\nabla\Delta N}^{INS} \\ Q_{b,\nabla\Delta N}^{INS T} & Q_{\nabla\Delta N}^{INS} \end{bmatrix} \\ &= (H^T Q_y^{-1} H)^{-1} \\ &= \begin{bmatrix} Q_{b_{KF}^-}^{-1} + \nabla G^T (Q_{\nabla\Delta\rho}^{-1} + Q_{\nabla\Delta\phi}^{-1}) \nabla G & \Lambda^T \nabla G^T Q_{\nabla\Delta\phi}^{-1} \\ \Lambda^T Q_{\nabla\Delta\phi}^{-1} \nabla G & \Lambda^T Q_{\nabla\Delta\phi}^{-1} \Lambda \end{bmatrix}^{-1} \end{aligned} \quad (\text{A-4})$$

where Q_b^{INS} and $Q_{\nabla\Delta N}^{INS}$ are the variance–covariance matrices of estimates of b and $\nabla\Delta N$, respectively. $Q_{b,\nabla\Delta N}^{INS}$ is the covariance matrix of the estimates.

By applying the inverse formula of the partitioned matrix and Woodbury matrix identity to Equation (A-4), $Q_{\nabla\Delta N}^{INS}$ can be expressed as:

$$Q_{\nabla\Delta N}^{INS} = \Lambda^{-T} \left\{ Q_{\nabla\Delta\phi} + \nabla G (Q_{b_{KF}^-}^{-1} + \nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G)^{-1} \nabla G^T \right\} \Lambda^{-1} \quad (\text{A-5})$$

The inverse formula of the partitioned matrix and Woodbury matrix identity [29] are, respectively, as follow:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{pmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{pmatrix} \quad (\text{A-6})$$

$$(D - CA^{-1}B)^{-1} = D^{-1} - D^{-1}C(-A + BD^{-1}C)^{-1}BD^{-1} \quad (\text{A-7})$$

where A and D are an arbitrary nonsingular square matrices, and B and C are arbitrary matrices.

APPENDIX B. A POSITIVE SEMI DEFINITE PROPERTY OF $Q_{\nabla\Delta N}^0 - Q_{\nabla\Delta N}^{INS}$

This appendix provides elements of the proof that the matrix obtained by subtracting $Q_{\nabla\Delta N}^{INS}$ from $Q_{\nabla\Delta N}^0$ is a positive semi-definite matrix. If the time-updated position from the INS is not used as the pseudo measurement (i.e., $Q_{b_{KF}}^{-1} \rightarrow \infty$), the variance-covariance matrix of the float ambiguity solution ($Q_{\nabla\Delta N}^0$) can be expressed as:

$$Q_{\nabla\Delta N}^0 = \Lambda^{-T} \{ Q_{\nabla\Delta\phi} + \nabla G \cdot (\nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G)^{-1} \cdot \nabla G^T \} \Lambda^{-1} \quad (\text{A-8})$$

by substituting $Q_{b_{KF}}^{-1} \rightarrow 0$ into Equation (A-5). Then, the difference between $Q_{\nabla\Delta N}^0$ and $Q_{\nabla\Delta N}^{INS}$ is given as:

$$Q_{\nabla\Delta N}^0 - Q_{\nabla\Delta N}^{INS} = \Lambda^{-T} \left[\nabla G \left\{ (\nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G)^{-1} - (Q_{b_{KF}}^{-1} + \nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G)^{-1} \right\} \nabla G^T \right] \Lambda^{-1} \quad (\text{A-9})$$

Since the inverse matrix of a positive definite matrix is also positive definite, $Q_{\nabla\Delta\rho}^{-1}$ and $Q_{b_{KF}}^{-1}$ are positive definite. From the definition of positive definite, we can show that $\nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G$ in Equation (A-9) is a positive definite by the following:

$$\begin{aligned} \forall v \in \mathbb{R} \setminus \{0\} \quad v^T \cdot (\nabla G^T Q_{\nabla\Delta\rho}^{-1} \nabla G) \cdot v &= (\nabla G \cdot v)^T \cdot Q_{\nabla\Delta\rho}^{-1} \cdot (\nabla G \cdot v) \\ &= y^T \cdot Q_{\nabla\Delta\rho}^{-1} \cdot y \quad \text{where } y = \nabla G \cdot v \in \mathbb{R} \setminus \{0\} \\ &> 0 \end{aligned} \quad (\text{A-10})$$

The last strict inequality follows because ∇G is a tall matrix whose number of rows is the same or greater than its columns. In this context, if the term in the braces on the right-hand side of Equation (A-9) is positive definite, we can show that $Q_{\nabla\Delta N}^0 - Q_{\nabla\Delta N}^{INS}$ is a positive-semi definite by the following:

$$\begin{aligned} \forall v \in \mathbb{R} \setminus \{0\} \quad v^T \cdot (\Lambda^{-T} \nabla G \cdot K \cdot \nabla G^T \Lambda^{-1}) \cdot v &= (\nabla G^T \Lambda^{-1} \cdot v)^T \cdot K \cdot (\nabla G^T \Lambda^{-1} \cdot v) \\ &= y^T \cdot K \cdot y \quad \text{where } y = \nabla G^T \Lambda^{-1} \cdot v \in \mathbb{R} \\ &\geq 0 \end{aligned} \quad (\text{A-11})$$

where K is a positive definite symmetric matrix. The last inequality follows because ∇G^T is a wide matrix whose number of columns is the same or greater than its rows, and Λ^{-1} is a full rank square matrix. Thus, in the following section, we will show that the term in the braces is a positive definite matrix.

Let us suppose that M and N are positive definite symmetric matrices. The inverse matrices of M and N exist, and are positive definite symmetric matrices. Then, we have the following:

$$\begin{aligned} M^{-1} - (M + N)^{-1} &= M^{-1} \{ (M + N) - M \} (M + N)^{-1} \\ &= M^{-1} N (M + N)^{-1} \\ &= M^{-1} N (I + M^{-1} N)^{-1} M^{-1} \\ &= M^{-1} (M^{-1} + N^{-1})^{-1} M^{-1} \end{aligned} \quad (\text{A-12})$$

From the definition of the positive definite, we can show that:

$$\begin{aligned}
\forall v \in \mathbb{R} \setminus \{0\} \quad v^T \cdot (M^{-1} + (M + N)^{-1}) \cdot v &= v^T \cdot M^{-1}(M^{-1} + N^{-1})^{-1}M^{-1} \cdot v \\
&= v^T \cdot M^{-T}(M^{-1} + N^{-1})^{-1}M^{-1} \cdot v \\
&= y \cdot (M^{-1} + N^{-1})^{-1} \cdot y \quad \text{where } y = M^{-1}v \in \mathbb{R} \setminus \{0\} \\
&> 0
\end{aligned} \tag{A-13}$$

The second equality follows from the fact that M^{-1} is symmetric. The last strict inequality follows because M^{-1} is a full rank square matrix. Since $\nabla G^T Q^{-1}_{\nabla \Delta \rho} \nabla G$ and $Q_{b_{KF}}^{-1}$ are positive definite and obviously symmetric, the term in the braces on the right-hand side of Equation (A-9) is a positive definite matrix. Therefore, from Equation (A-9), (A-11), and (A-13), $Q_{\nabla \Delta N}^0 - Q_{\nabla \Delta N}^{INS}$ is positive semi-definite.

APPENDIX C. PROBABILITY OF CORRECT FIX WRT. THE AMBIGUITY PRECISION

In this appendix, we will show that the PCF computed from the variance–covariance matrix Q_1 is the same or greater than that of Q_2 , when $Q_2 \geq Q_1$ (i.e. $Q_2 - Q_1$ is positive semi-definite). Let the triangular decompositions of Q_1 and Q_2 be given as $Q_1 = L_1 D_1 L_1^T$ and $Q_2 = L_2 D_2 L_2^T$, respectively. Since L_2 is full-rank low triangular matrix, we have the following:

$$\begin{aligned}
\forall v \in \mathbb{R} \quad v^T \cdot (D_2 - L_2^{-1} Q_1 L_2^{-T}) \cdot v &= v^T \cdot (L_2^{-1} Q_2 L_2^{-T} - L_2^{-1} Q_1 L_2^{-T}) \cdot v \\
&= v^T \cdot (L_2^{-1} (Q_2 - Q_1) L_2^{-T}) \cdot v \\
&= y^T \cdot (Q_2 - Q_1) \cdot y \quad \text{where } y = L_2^{-T} v \in \mathbb{R} \\
&\geq 0
\end{aligned} \tag{A-14}$$

The last inequality follows because $Q_2 - Q_1$ is positive semi-definite. Thus, the diagonal entries of D_2 are the same or greater than those of $L_2^{-1} Q_1 L_2^{-T}$. Furthermore, let define the triangular decomposition of $L_2^{-1} Q_1 L_2^{-T}$ as $L_3 D_3 L_3^T$. From the property of the triangular decomposition, the diagonal entries of $L_2^{-1} Q_1 L_2^{-T}$ are the same or greater than those of D_3 . Thus, we can construct the following inequalities:

$$\begin{aligned}
\text{diag}(D_2) &\geq \text{diag}(L_2^{-1} Q_1 L_2^{-T}) \\
&\geq \text{diag}(D_3)
\end{aligned} \tag{A-15}$$

where $\text{diag}(\cdot)$ represents the diagonal components of the matrix, and $\geq$ denotes an element-wise operation.

From the definition of Q_1 , it can be expressed as:

$$\begin{aligned}
Q_1 &= L_1 D_1 L_1^T \\
&= L_2 L_3 \cdot D_3 \cdot (L_2 L_3)^T
\end{aligned} \tag{A-16}$$

Since L_2 and L_3 are lower unit triangular matrices and the triangular decomposition of a matrix is unique, the Equation (A-16) implies that D_1 is the same as D_3 . Therefore, we have that $\text{diag}(D_2) \geq \text{diag}(D_1)$. This shows that $Q_2 \geq Q_1$ implies that the diagonal entries of D_2 representing the conditional least-squared variance of Q_2 are the same or greater than their counterparts of Q_1 . This proves that the PCF computed from Q_1 is the same or greater than that of Q_2 , when $Q_2 \geq Q_1$.

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